Introduciton to mobile Robotics

PROJECT – 2026

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Project Defintion

GOALS AND REQUIREMENTS

Desired Behaviour

Initial Boot & Undocking

Autonomous Exploration

Proportional Obstacle Avoidance:

- AWARE zone: slowdown. WARNING zone: braking CLOSE zone: stop and turn

Hazard Recovery:

- Recover from bumper contact with obstacle

Timed Return-to-Dock

Human Override:

- TELEOP for manual control

Architecture

DESIGN DECISIONS

```
┌─────────────────────────────┐
│  KeyboardHandler Node       │
├─────────────────────────────┤
│ Publishes:                  │
│   /mode (String)            │
│   /cmd_vel (Twist)          │
└─────────────────────────────┘
             │
  ┌──────────────────────────────┼───────────────────────────────┐
  │                      │                        │
  ▼                      ▼                        ▼
/mode ─────────────► [Explorer Node] ◄──────────────── /cmd_vel (manual)
                    │
                    │ Subscriptions:
                    │   /ir_intensity
                    │   /hazard_detection
                    │   /dock_status
                    │   /mode
                    │
                    │ Publishes:
                    │   /cmd_vel (autonomous)
                    │   /cmd_lightring
                    │
                    │ Action Clients:
                    │   undock
                    │   dock
                    │
                    ▼
          ┌────────────────────────────────┐
          │  Robot Motion Controller       │
          │  (Create 3 base)               │
          └────────────────────────────────┘
```

Design Decisions

2 Node achitecture:
- Manual control
- Exploration and obstactle avoidance

Communication via topics
- Subsciber and publisher
- Organisation of communication
- Callback methods on message receive

Design Decisions

Global State of the Explorer

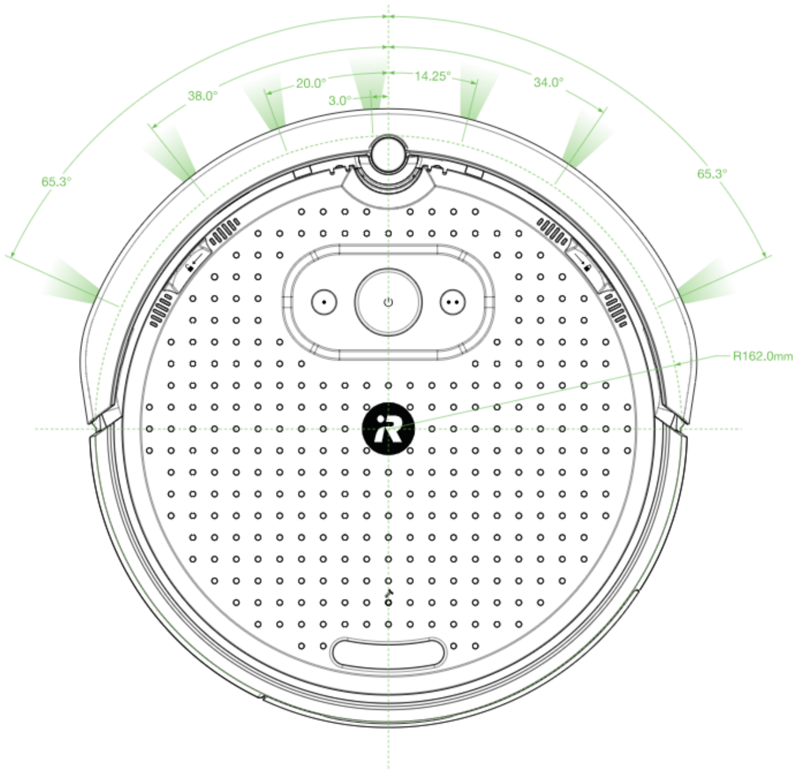
- Easy conditional Behaviour
- Extendable framework for additional features

```
DOCKED: Robot is on the charging dock (initial state)
UNDOCKING: Robot is in process of leaving the dock
EXPLORING: Robot is actively wandering and avoiding obstacles
ESCAPING: Robot disengages form bumped obstacle
RETURNING: Robot has completed mission and is returning to dock
MANUAL: Human has taken manual control via override
COMPLETED: Robot gracefully completed mission (final state)
```

Implementation

Obstacle Avoidance

- Subscribe to ir-sensor topic
- On message arrival, execute obstacle check:
  - 7 different sensors
  - Check returns against threshold
  - Determine direction from sensor position
- On detection initiate avoidance



Manual Control

keyboard logger running constantly

On interrupt:

- sent message to change status of the explorer node

In manual mode:

- Read commands from keystrokes
- Relay commands to robot

Return to automatic:

- Change mode of explorer node

Return to Dock

Final docking handled by docking server

- Only works in proximity to Station

Navigate back to Dock

- Explore Until Dock is detected
- Keep obstacle avoidance active on way to dock
- Seperate state to keep tack of docking goal

Manual mode for accelerated docking

LED Status

Indicate Status of the Robot

Color	Meaning
Green	Path clear / nominal autonomous motion
Yellow	Early obstacle awareness / gentle veer
Orange	Warning zone / braking + stronger turn
Red	Critical close obstacle / no forward + aggressive turn
Purple	Bumper recovery (`ESCAPING`)
Blue	Manual control (`TELEOP`)

Results

DEMONSTRATION

Anyone got a video?

Sources

The pictures are taken form the create3 website: Create® 3 Docs